PART 1 · Probability and random signals

U1 · Total probability and Bayes

IF YOU SEE "machine M_i produces 40 % … defective 2 % for M_1 … probability that a chip is defective / given that it is defective, which machine": sources with shares and rates.

1. Shares $P(A_i)$: percentages, or counts divided by the total (18 of 55 patients → 18/55). They add to 1.
2. Rate of the asked event in each source $P(B \mid A_i)$. If the text gives the opposite event (85 % pass, you want "not pass"), use $1 - 0.85$.
3. Forward ("probability that an item is B"): $P(B) = \sum_i P(B \mid A_i)P(A_i)$; write each product, then add.
4. Backward ("given that it is B, probability it came from A_k"): $P(A_k \mid B) = \dfrac{P(B \mid A_k)P(A_k)}{P(B)}$: that source's product over the sum. All backward answers add to 1.

Jun26 1b: $0.4(0.02) + 0.35(0.04) + 0.25(0.05) = 0.0345$; $P(M_2 \mid D) = 0.014/0.0345 = 0.41$. *Apr24 1b (TB, "99 % accurate" = 0.99 both ways):* $0.000495/0.01049 = 0.047$. *Lifetime* $\Pr[T \ge t] = e^{-t/5}$, *"breaks in year 3 given alive at 2":* $(e^{-2/5} - e^{-3/5})/e^{-2/5} = 1 - e^{-1/5} = 0.18$.

Independent events: $P(A \cap B) = P(A)P(B)$, i.e. $P(A \mid B) = P(A)$. Not "$P(A \cap B) = 0$" (disjoint) and not "$P(A \cup B) = P(A) + P(B)$".

U2 · Joint pdf: constant, marginal, conditional, independence

IF YOU SEE "joint PDF $p_{X,Y}(x,y) = \ldots$ for $0 \le x \le 1$, $0 \le y \le 1$ (square) or $x + y < 1$ / $0 \le y \le x \le 1$ (triangle)": limits are the whole job.

1. Draw the region. Square: both limits are numbers. Triangle: choose the order, freeze the outer letter, inner limits depend on it (x from 0 to $1 - y$), outer limits are numbers (y from 0 to 1).
2. Constant c: double integral = 1. Inner antiderivative, evaluate at both limits, then outer. *Apr25 1c:* $\int_0^1 \int_0^{1-y} (cx + 1)\,dx\,dy = \frac{c}{6} + \frac{1}{2} = 1 \Rightarrow c = 3$.
3. Marginal $p_X(x) = \int p_{X,Y}\,dy$ with the y limits for that fixed x; always write the domain ("for $0 \le x \le 1$"). *Jun26 1c:* $\int_0^1 6x^2 y\,dy = 3x^2$. *Apr24 1c:* $p_Y(y) = \int_y^1 2\,dx = 2(1 - y)$.
4. Conditional $p_{X|Y}(x \mid y) = p_{X,Y}/p_Y(y)$, domain = the x range for that y. *Apr24 1c:* $2/(2(1 - y)) = 1/(1 - y)$ *for* $y \le x \le 1$.
5. Independent only if the region is a rectangle AND the formula factors ($6x^2 y = 3x^2 \cdot 2y$: yes; anything on a triangle: no).
6. Discrete table: marginal of X = add across the row ($\frac{1}{6} + \frac{1}{4} + \frac{1}{8} = \frac{13}{24}$).

U3 · Random variables and processes (MC facts)

IF YOU SEE "WSS", "ergodic", "expected value", "discrete or continuous", "realisation", "legitimate autocorrelation":

- Expected value = average of the outcomes weighted by the pmf/pdf = mean over infinitely many observations. Not the most probable, not the largest.
- Discrete = countable values (counts, phone numbers, pixel 0–255). Continuous = any value in a range (time, weight).
- A picture is a realisation of a random process; one pixel is a random variable. The ensemble is all realisations.
- WSS: mean constant in time AND autocorrelation depends only on the lag $t - s$ (constant pdf = strict-sense; flat PSD = white noise; time averages = ergodicity).
- Ergodic in the mean: the time average of ONE realisation equals the ensemble mean. i.i.d. dice rolls: ergodic in mean and variance. A random amplitude drawn once per realisation ($A\cos t$, uniform on $[-A, A]$): not ergodic in variance; $A\cos t$ is not even WSS (mean $\cos t$ moves); uniform on $[-A, A]$ has mean 0 and is ergodic in the mean.
- Recipe (Apr24 1d): (1) $E[X(t)]$ over the random ingredient: constant? (2) $E[X(t)X(s)]$ with $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$ (terms with $2U$ average to 0): function of $t - s$ only? (3) time average of one realisation equals (1)? $\cos(t + U)$: 0, $\frac{1}{2}\cos(t - s)$, 0 → WSS and ergodic.
- Legitimate autocorrelation: $r[-l] = r[l]$ and $|r[l]| \le r[0]$..
- Covariance: direction by its sign, scale-dependent, unbounded ("all of the above"); correlation coefficient = $\mathrm{cov}/(\sigma_X\sigma_Y)$ in $[-1, 1]$. Binomial "exactly k of n": $\binom{n}{k}p^k(1-p)^{n-k}$; $\binom{8}{6}0.8^6 0.2^2 = 0.2936$.

U4 · Spectral factorisation (PSD → $L(z)$, σ_i^2)

IF YOU SEE "find the spectral factorization / the innovation filter $L(z)$ and the variance of the input noise" with $P(e^{j\theta}) = \dfrac{\text{number}}{\alpha \pm \beta\cos\theta}$:

1. $\cos\theta = \frac{z+z^{-1}}{2}$, so $\alpha + \beta\cos\theta = \alpha + \frac{\beta}{2}z + \frac{\beta}{2}z^{-1}$.
2. Match to $C(1 \pm az^{-1})(1 \pm az) = C(1 + a^2) \pm Ca(z + z^{-1})$ (sign + if $\beta > 0$, − if $\beta < 0$): $Ca = |\beta|/2$ and $C(1 + a^2) = \alpha$.
3. Divide: $|\beta|a^2 - 2\alpha a + |\beta| = 0$; $a = \dfrac{2\alpha \pm \sqrt{4\alpha^2 - 4\beta^2}}{2|\beta|}$. Keep the root with $|a| < 1$ (minimum phase; the other is $1/a$). $C = |\beta|/(2a)$.
4. $\sigma_i^2 = C_{\text{numerator}}/C_{\text{denominator}}$; $L(z)$ = the z^{-1} halves only: $\dfrac{1 \pm bz^{-1}}{1 \pm az^{-1}}$.
5. Check by expanding $\sigma_i^2|L(e^{j\theta})|^2 = \sigma_i^2 \frac{1+b^2\pm 2b\cos\theta}{1+a^2\pm 2a\cos\theta}$ back.

Apr25 1e: $1/(61 + 60\cos\theta)$: $Ca = 30$, $C(1 + a^2) = 61$, $30a^2 - 61a + 30 = 0$, $a = 5/6$ *(not 1.2)*, $C = 36$: $\sigma_i^2 = 1/36$, $L = 1/(1 + \frac{5}{6}z^{-1})$. *Apr24 1e:* $7/(25 - 24\cos\theta)$: $12a^2 - 25a + 12 = 0$, $a = 3/4$, $C = 16$: $\sigma_i^2 = 7/16$, $L = 1/(1 - \frac{3}{4}z^{-1})$. *Nov22 1d:* $1/(\frac{5}{3} - \cos\theta)$: $a = \frac{1}{3}$, $C = \frac{3}{2}$: $\sigma_i^2 = \frac{2}{3}$, $L = 1/(1 - \frac{1}{3}z^{-1})$. MC: the innovation filter is minimum phase (poles and zeros inside the unit circle), causal, stable; $L(z^{-1})$ (zeros outside) is the wrong mirror; the constant in front of the factorised PSD is σ_i^2 (Jan22 1d: 2, option c).

PART 2 · Estimation

U5 · Maximum likelihood (MLE)

IF YOU SEE "write the likelihood and log-likelihood … derive the MLE for θ", "find an estimator for the parameter" with a pdf $p(x;\theta)$ and N i.i.d. samples:

1. Likelihood = product of the pdf over the samples: constants become c^N, exponentials merge: $\exp(\sum \cdot)$, θ^k in a denominator becomes θ^{Nk}.
2. ln of the product = sum of lns: $\ln(ab) = \ln a + \ln b$, $\ln a^n = n\ln a$, $\ln e^u = u$, denominator terms get a minus.
3. Drop every term without θ.
4. Differentiate in θ: $\frac{d}{d\theta}c\ln\theta = c/\theta$; $\frac{d}{d\theta}(c/\theta) = -c/\theta^2$; $\frac{d}{d\theta}(c/\theta^2) = -2c/\theta^3$; $\frac{d}{d\theta}c\theta = c$; nested $\lambda(\theta)$: $\frac{d}{d\theta}\ln\lambda = \lambda'/\lambda$ (sigmoid: $\lambda' = \lambda(1 - \lambda)$ so $\lambda'/\lambda = 1 - \lambda$).
5. Set to zero, multiply by the denominators, solve for θ. Discard solutions at the boundary ($\lambda = 1$ means $\theta = \infty$). If the question's parameter is inside a sigmoid, invert at the end: $\frac{1}{1+e^{-\theta}} = v \Rightarrow \theta = -\ln(\frac{1}{v} - 1)$.
6. One sentence of meaning ("the sample mean divided by k").

Gamma, known k (Jun26 2b): $\ln L = (k-1)\sum\ln x_i - \frac{\sum x_i}{\theta} - nk\ln\theta - n\ln\Gamma(k)$; $\frac{S}{\theta^2} - \frac{nk}{\theta} = 0 \Rightarrow \hat\theta = \frac{\sum x_i}{nk} = \bar x/k$. $\frac{x}{2\theta}e^{-x^2/(2\theta)}$ *(Apr24 2c):* $\hat\theta = \frac{1}{2N}\sum x_n^2$. *Rayleigh* $\frac{x}{\theta^2}e^{-x^2/(2\theta^2)}$ *(Nov19 3a):* $\hat\theta = \sqrt{\frac{1}{2N}\sum x^2}$. $\lambda e^{-\lambda x}$: $\hat\lambda = N/\sum x_i$. *Sigmoid (Apr25 2c)* $3\lambda e^{-4\lambda x/7}$: $(1 - \lambda)(1 - \frac{4}{7}\lambda x) = 0 \Rightarrow \lambda = \frac{7}{4x} \Rightarrow \hat\theta = -\ln(\frac{4x}{7} - 1)$. *Angle (Nov22 2d):* $\hat\theta = \tan^{-1}(x_2/x_1)$, *quadrant of* (x_1, x_2).

U6 · Posterior and MAP

IF YOU SEE "assume a Beta/Gamma/exponential prior … write the likelihood, derive the posterior, compute the MAP, compare with the MLE":

1. Likelihood with the data numbers in ($\sum x_i$, n). Prior with constants dropped: Beta$(\alpha,\beta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$; Gamma$(\alpha,\beta) \propto \lambda^{\alpha-1}e^{-\beta\lambda}$.
2. Posterior ∝ likelihood × prior: add exponents of θ and of e. Name it: $\theta^{a-1}(1-\theta)^{b-1}$ is Beta(a,b); $\lambda^{a-1}e^{-b\lambda}$ is Gamma(a,b). The denominator $p(x)$ is a number: ignore it for the MAP.
3. MAP = peak: ln, differentiate, zero, solve. Beta: $\frac{a-1}{\theta} = \frac{b-1}{1-\theta} \Rightarrow \hat\theta = \frac{a-1}{a+b-2}$. Gamma: $\frac{a-1}{\lambda} = b \Rightarrow \hat\lambda = \frac{a-1}{b}$.
4. MLE = same without the prior. The prior pulls the MAP toward its own peak; the pull fades as n grows (data terms grow, prior terms do not).

NegBin $k = 2$, Beta(2,3), $x = \{3, 5, 4, 6, 4\}$ *(Jun26 2c):* $L = \prod(x_i - 1)\,\theta^{10}(1-\theta)^{12}$ ($\sum x_i = 22$, $\sum(x_i - 2) = 12$); *posterior* $\theta^{11}(1-\theta)^{14}$ *= Beta(12,15); MAP* $= 11/25 = 0.44$ *(MLE* $10/22$*, prior peak* $1/3$*). Poisson, Gamma(3,2),* $x = \{2, 4, 3, 5, 2, 4\}$ *(Jun26 2d):* $L \propto \lambda^{20}e^{-6\lambda}$; *posterior* $\lambda^{22}e^{-8\lambda}$ *= Gamma(23,8); MAP* $22/8 = 2.75$; *MLE* $20/6 = 3.33$; *the prior (mode 1) pulls down. Posterior given (Apr25 2d)* $\theta^A(1-\theta)^B$: *MAP* $= A/(A + B)$ *with* $A = \beta_1 - 1 + k_1N_1$, $B = \beta_0 - 1 + k_0N_0$. *Nov22 2g* $(x + a)^2\theta e^{-(x+a)\theta}$: $\frac{1}{\theta} - (x + a) = 0 \Rightarrow \frac{1}{x+a}$. *Sigmoid posterior (Apr24 2d)* $\frac{\lambda}{2}e^{-\lambda x/2}$: $(1 - \lambda)(1 - \frac{x\lambda}{2}) = 0 \Rightarrow \lambda = \frac{2}{x} \Rightarrow \hat\theta = -\ln(\frac{x}{2} - 1)$, $x > 2$. *No closed form (Nov19 3c): stop at the equation "derivative* $= 0$*". Discrete* $X \in \{\pm 1\}$, $Y = X + N$ *(Jan22 2f, 2h):* $P[X = 1 \mid Y] = \frac{e^{-(y-1)^2/2\sigma^2}}{e^{-(y-1)^2/2\sigma^2} + e^{-(y+1)^2/2\sigma^2}}$ *(a sum in the denominator); ratio* $e^{2y/\sigma^2} \gtrless 1 \Rightarrow y_{th} = 0$.

U7 · MMSE = posterior mean

IF YOU SEE "find the minimum mean square error estimator" (posterior given or derived in the previous part):

1. $\hat\theta_{MMSE} = E[\theta \mid x] = \int \theta\, p(\theta \mid x)\,d\theta$ over the posterior's range; the posterior must integrate to 1 (divide likelihood × prior by ∫ likelihood × prior if needed). Discrete: $\sum_x x\,P[X = x \mid Y]$.
2. Substitute to match the handed-out integral ($y = (x + a)\theta$: $\theta = y/(x + a)$, $d\theta = dy/(x + a)$), evaluate antiderivative at ∞ (0) and at 0, subtract.

Nov22 2f: $(x + a)^2\int_0^\infty \theta^2 e^{-(x+a)\theta}d\theta = \frac{1}{x+a}\int_0^\infty y^2 e^{-y}dy = \frac{2}{x+a}$ *(MAP was* $\frac{1}{x+a}$*: skewed posterior). Jan22 2g:* $(+1)P[1 \mid Y] - P[-1 \mid Y] = \tanh(y/\sigma^2)$. *Mixture* $c\,\mathcal{N}(x, 1) + (1 - c)\mathcal{N}(-x, 1)$: $x(2c - 1)$. Means: Beta(a,b): $\frac{a}{a+b}$; Gamma(a,b): $\frac{a}{b}$; exponential rate r: $\frac{1}{r}$. MAP = MMSE only for symmetric posteriors (Gaussian); "always different" is false.

U8 · Fisher information and CRLB

IF YOU SEE "derive the Fisher information", "find the Cramér–Rao lower bound", "check whether the CRLB can be derived":

1. If the pdf's range depends on θ ($0 \le x \le \theta$): regularity condition violated, no CRLB. Stop (Jan22 2e).
2. $\ln p$ of the full data (θ terms only). 2. Differentiate twice. 3. Expectation: replace x, x^2 by $E[x]$, $E[x^2]$ (given in a hint, or integrate). 4. $\mathcal{I}(\theta) = -E[\partial^2 \ln p/\partial\theta^2]$; N samples: $\mathcal{I} = N\,i(\theta)$; CRLB $= 1/\mathcal{I}$.

Nov22 2e: $\ln p \ni \frac{A}{\sigma^2}(x_1\cos\theta + x_2\sin\theta)$; *second derivative* $\frac{A}{\sigma^2}(-x_1\cos\theta - x_2\sin\theta)$; $E[x_1] = A\cos\theta$, $E[x_2] = A\sin\theta \rightarrow \mathcal{I} = A^2/\sigma^2$. *Rayleigh (Nov19 3b):* $\ln p = \ln x - 2\ln\theta - \frac{x^2}{2\theta^2}$; *second derivative* $\frac{2}{\theta^2} - \frac{3x^2}{\theta^4}$; $E[x^2] = 2\theta^2 \rightarrow i = 4/\theta^2$, *CRLB* $= \theta^2/(4N)$. MC: Var $\ge 1/\mathcal{I}$ (not $\ge \mathcal{I}$); $\mathcal{I} \ge 0$ (never "strictly negative"); $\mathcal{I}_N = Ni$ so the bound falls as $1/N$; efficient = unbiased AND variance equal to the bound; MLE is asymptotically unbiased/efficient/Gaussian, not "always" at the bound; invariance $\hat\theta = \psi^{-1}(\hat\alpha)$ holds for ML and LS, not MMSE.

U9 · Least squares

IF YOU SEE "find the linear function having least squares error for the data $\{(x_i, y_i)\}$":

$J(a,b) = \sum(y_i - ax_i - b)^2$; $\partial J/\partial a = -2\sum x_i(y_i - ax_i - b) = 0$, $\partial J/\partial b = -2\sum(y_i - ax_i - b) = 0$; solve the 2×2 system by substitution, fractions kept. Closed form: $a = \frac{n\sum xy - \sum x\sum y}{n\sum x^2 - (\sum x)^2}$, $b = \frac{\sum y - a\sum x}{n}$. *Apr25 2b,* $(0,3), (2,4), (3,3)$: $26a + 10b = 34$, $10a + 6b = 20 \rightarrow a = \frac{1}{14}$, $b = \frac{45}{14}$. "Which estimator, no noise pdf given": least squares (needs no statistics). Linear in the unknown (ϕ known in $Ae^{\phi n}$): closed form $\hat A = \sum x[n]e^{\phi n}/\sum e^{2\phi n}$. Non-linear: write the cost, iterate (Gauss–Newton or Newton–Raphson) or transform ($x = \ln\theta + w \Rightarrow \hat\theta = e^{\hat x}$). The error vector, not the solution, is orthogonal to the columns of $\mathbf{H}$.

U10 · Estimation MC: the swap test

A false option is a true sentence with one word swapped: *always · regardless · only · strictly* · $x \leftrightarrow \theta$ · *mean* $\leftrightarrow$ *maximum* · $\mathcal{I} \leftrightarrow 1/\mathcal{I}$ · *negative*. True: MLE = $\arg\max_\theta p(x \mid \theta)$ ($\arg\min\sum(\cdot)^2$ is LS, $\arg\max p(\theta \mid x)$ is MAP); MMSE = posterior mean; MAP = posterior maximum, needs a prior; Bayes risk $\iint C(e)p(\theta \mid x)d\theta\,p(x)dx$ is minimised: quadratic cost → MMSE, hit-or-miss → MAP; Bayesian estimators DO use prior knowledge.

PART 3 · Spectral estimation

U11 · Correlogram from lags

IF YOU SEE "calculate the correlogram / the PSD in a non-parametric fashion … as a function of cosines" with $r[0], r[\pm1], r[\pm2]$ and "assume $r[k] = 0$ beyond":

$\hat P_{corr}(e^{j\theta}) = r[0] + 2r[1]\cos\theta + 2r[2]\cos 2\theta + \cdots$ (every lag except 0 is doubled; comes from $r[k]e^{-jk\theta} + r[-k]e^{jk\theta} = 2r[k]\cos k\theta$). Values: at $\theta = 0$ all cosines are 1; at π they alternate $-1, +1, -1$; at $\pi/2$: $0, -1, 0, +1$. It may go negative (a flaw of truncation); a parametric PSD never does. *Jun26 3c* $(3, 2, 1)$: $3 + 4\cos\theta + 2\cos 2\theta$ *(9 at 0, 1 at* π*). Apr25 3d* $(2, -\frac{1}{4}, -\frac{3}{16})$: $2 - \frac{1}{2}\cos\theta - \frac{3}{8}\cos 2\theta$ *(1.125 at 0, 2.125 at* π*). Apr24 3d:* $\frac{25}{16} - \frac{3}{2}\cos\theta + \cos 2\theta - \cos 3\theta$ *(0.06 at 0, 5.06 at* π*). Jan22 3c:* $\frac{13}{8} - \frac{15}{8}\cos\theta + \frac{1}{2}\cos 2\theta$.

U12 · AR model by Yule–Walker

IF YOU SEE "model as an AR(p) process, find the parameters / state the Yule–Walker equations in matrix form / variance of the driving noise / PSD as a function of $\cos\theta$":

Model (course signs): $x[n] = i[n] - a_1x[n-1] - a_2x[n-2]$, $H(z) = 1/(1 + a_1z^{-1} + a_2z^{-2})$.

1. Yule–Walker, one line per lag $m = 1..p$: $r[m] + a_1r[m-1] + a_2r[m-2] = 0$ (use $r[-1] = r[1]$). Matrix:

$\begin{bmatrix} r_0 & r_1 \\ r_1 & r_0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = -\begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$ (or augmented $\begin{bmatrix} r_0 & r_1 & r_2 \\ r_1 & r_0 & r_1 \\ r_2 & r_1 & r_0 \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sigma_i^2 \\ 0 \\ 0 \end{bmatrix}$). AR(1): $a_1 = -r[1]/r[0]$.

2. $\sigma_i^2 = r[0] + a_1 r[1] + a_2 r[2]$ (must be between 0 and $r[0]$).
3. $P_{AR} = \dfrac{\sigma_i^2}{|1 + a_1 e^{-j\theta} + a_2 e^{-j2\theta}|^2}$; AR(1) denominator $1 + a_1^2 + 2a_1 \cos\theta$; AR(2) denominator $(1 + a_1^2 + a_2^2) + 2(a_1 + a_1 a_2)\cos\theta + 2a_2 \cos 2\theta$. Clear fractions by multiplying top and bottom.

Jun26 3d $(3, 2, 1)$: $3a_1 + 2a_2 = -2$, $2a_1 + 3a_2 = -1 \to a_1 = -0.8$, $a_2 = 0.2$, $\sigma_i^2 = 3 - 1.6 + 0.2 = 1.6$, $P = 1.6/(1.68 - 1.92\cos\theta + 0.4\cos 2\theta)$ *(10 at 0, 0.4 at* π*). Apr25 3e* $(2, -\frac{1}{4})$*:* $a_1 = \frac{1}{8}$, $\sigma_i^2 = 2 - \frac{1}{32} = \frac{63}{32}$, $P = \frac{63/32}{65/64 + \frac{1}{8}\cos\theta} = \frac{126}{65 + 16\cos\theta}$. *Nov22 3f* $(1, -\frac{1}{8})$*:* $a_1 = \frac{1}{8}$, $\sigma_i^2 = \frac{63}{64}$, $P = \frac{63}{65 + 16\cos\theta}$. *Jan22 3d* $(\frac{13}{8}, -\frac{15}{16})$*:* $a_1 = \frac{15}{26}$, $\sigma_i^2 = \frac{451}{416}$. *Apr24 3e* $(\frac{25}{16}, -\frac{3}{4}, \frac{1}{2})$*:* $a_1 = \frac{204}{481}$, $a_2 = -\frac{56}{481}$, $\sigma_i^2 = 1.19$.

U13 · MA model from lags

IF YOU SEE "model as an MA(1) process, find the parameters … PSD as a function of cosines", "MA(2) … calculate a and b":

MA(1) $x = i + b_1 i[n-1]$: $r[0] = \sigma_i^2(1 + b_1^2)$, $r[1] = \sigma_i^2 b_1$, $r[k \geq 2] = 0$.

1. $\rho = r[1]/r[0] = \dfrac{b_1}{1 + b_1^2} \to \rho b_1^2 - b_1 + \rho = 0 \to b_1 = \dfrac{1 \pm \sqrt{1 - 4\rho^2}}{2\rho}$; the roots multiply to 1; keep $|b_1| < 1$ (minimum phase).
2. $\sigma_i^2 = r[1]/b_1$ (check $\sigma_i^2(1 + b_1^2) = r[0]$).
3. $P_{MA} = \sigma_i^2(1 + b_1^2 + 2b_1\cos\theta) = r[0] + 2r[1]\cos\theta$ (ignores $r[2]$ and beyond). If the process is truly MA(1) (only lags $0, \pm 1$ given), this equals the correlogram exactly (Jan20 3b).

MA(2) with unit-variance input, $x = i + ai_1 + bi_2$: $r[2] = b$, $r[1] = a(1 + b)$, $r[0] = 1 + a^2 + b^2$. *Jan22 3b:* $b = \frac{1}{4}$, $a = -\frac{15}{16}/\frac{5}{4} = -\frac{3}{4}$, $r[0] = \frac{13}{8}$. *Apr25 3f and Nov22 3g:* $\rho = -\frac{1}{8} \to b_1^2 + 8b_1 + 1 = 0 \to b_1 = -4 + \sqrt{15} = -0.127$*;* $\sigma_i^2 = -0.25/-0.127 = 1.97$ *(Apr25) or* 0.984 *(Nov22);* $P = 2 - 0.5\cos\theta\,/\,1 - 0.25\cos\theta$. MC: MA fit needs non-linear equations (Jan19 1d C true); MA(q) autocorrelation is zero beyond lag q; linear prediction (Wiener) gives AR, not MA, coefficients.

U14 · Difference equation → model name, PSD, autocorrelation

IF YOU SEE "$x[k] = i[k] - i[k-1] + \frac{3}{4}x[k-1]$ … which kind of process / PSD / autocorrelation" or "filtered with $H(z) = \ldots$":

- Name: only i terms → MA(q), q = oldest kick; $i[n]$ + past x → AR(p); both → ARMA(p, q). $H(z)$: numerator = i coefficients, denominator = 1− (coefficients of past x on the right side).
- PSD $= \sigma_i^2|H(e^{j\theta})|^2$; $|1 + ce^{-j\theta}|^2 = 1 + c^2 + 2c\cos\theta$ top and bottom. *Nov22 3d:* $\frac{7}{8} \cdot \frac{2 - 2\cos\theta}{\frac{25}{16} - \frac{1}{2}\cos\theta} = 28\frac{1 - \cos\theta}{25 - 24\cos\theta}$ *(0 at* $\theta = 0$*).*
- MA lags: slide the coefficient list $[1, b_1, b_2, \ldots]$ over itself by k, add the products, times σ_i^2. *Apr24 3c,* $[1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{2}]$*:* $r_0 = \frac{25}{16}$, $r_1 = -\frac{1}{2} - \frac{1}{8} - \frac{1}{8} = -\frac{3}{4}$, $r_2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, $r_3 = -\frac{1}{2}$.
- ARMA(1,1) $x = i + b_1 i_1 - a_1 x_1$ (here $b_1 = -1$, $-a_1 = \frac{3}{4}$): multiply by x and by x_1, take E. Rules: $E[i^2] = \sigma_i^2$; $E[i\,x_1] = 0$ (future kick); $E[i_1 x_1] = \sigma_i^2$ (the past signal contains the past kick with coefficient 1); $E[x_1^2] = r_0$. *Nov22 3e:* $r_0 = \frac{7}{8} + \frac{7}{8} + \frac{9}{16}r_0 - \frac{3}{2}\cdot\frac{7}{8} \Rightarrow r_0 = 1$*;* $r_1 = -\frac{7}{8} + \frac{3}{4}r_0 = -\frac{1}{8}$*;* $r[\tau] = \frac{3}{4}r[\tau - 1]$ *for* $\tau \geq 2$*, so* $r[\tau] = -\frac{1}{8}(\frac{3}{4})^{|\tau|-1}$.

U15 · Spectral estimation theory

IF YOU SEE "which statement is TRUE/FALSE about the periodogram / windows / zero padding", "fill the table of PSD estimators", "energy or power signal", "Welch: how many segments", "AR model order":

Method	Idea	Bias	Variance
Periodogram (= correlogram, FT of the biased $\hat{r}$)	$\frac{1}{N}\lvert\sum x[n]e^{-jn\theta}\rvert^2$	asymptotically unbiased (FT of the biased $\hat{r}$)	$\propto P^2$, does not fall with N: NOT consistent
Bartlett	K non-overlapping segments, average periodograms	increases (shorter segments)	$/K$
Welch (WOSA)	overlapping (50 %) windowed segments, average	like Bartlett	roughly $/K$ but not exactly (correlated segments); lower than Bartlett for the same N
Blackman–Tukey	window the autocorrelation (lag window	more with shorter M	less; consistent
	$M < N$), then FT		

- Consistent: Bartlett, Welch, Blackman–Tukey, parametric. FT of autocorrelation: correlogram, Blackman–Tukey. Averages periodograms: Bartlett, Welch. Overlap: Welch. "Variance ∝ square of the true spectrum": periodogram (if the table says "bias", the intended answer is still periodogram).
- Welch segments: $K = \frac{N-L}{L(1-\text{overlap})} + 1$; variance ratio $1/K$ if independent. *Jun26 3b:* $N = 1000$, $L = 200$, *50 %:* $K = 800/100 + 1 = 9$, *ratio* $1/9$. Bartlett: $K = N/L$.
- Windows: main-lobe width → resolution (wider = worse); side-lobe height → leakage (higher = more). Same length: rectangular = narrowest main lobe (best resolution), highest side lobes (most leakage); triangular/Hanning/Hamming widen the lobe, lower the side lobes. Longer window → better resolution, leakage (relative side-lobe level) unchanged. Zero padding to L points = sampling the same FTD at $\theta = 2\pi k/L$: interpolation, NO resolution gain, no variance change.
- Jan22 3a: infinite cosine → two lines (B); windowed → sinc with side lobes (D); zero-padded DFT → circles sampling D (C); 24-point DFT at $f_s/2$ → only the peaks (A): 1B 2D 3C 4A.
- Energy $E = \int |x|^2 dt$ or $\sum |x[n]|^2$; power $P = \lim \frac{1}{T}\int_{-T/2}^{T/2} |x|^2 dt$. Finite E → energy signal, $P = 0$. Periodic → power signal, $E = \infty$. $A\sin\omega t$: $P = A^2/2$ *(via* $\sin^2 = \frac{1-\cos 2\omega t}{2}$*).* $e^{-n}u[n]$: $E = \sum(e^{-2})^n = \frac{1}{1-e^{-2}} = 1.157$. $e^{-t}u(t)$: $E = \frac{1}{2}$*;* $\cos t\,u(t)$: $P = \frac{1}{4}$.
- AR order (Jun26 3e): residual variance always decreases with p → cannot choose p; risk = overfitting (spurious peaks, parameter variance). Criteria add a penalty growing with p: AIC $= N\ln\hat{\sigma}_p^2 + 2p$, MDL $= N\ln\hat{\sigma}_p^2 + p\ln N$, FPE $= \hat{\sigma}_p^2\frac{N+p+1}{N-p-1}$; pick the minimum. Any spectrum (even MA nulls) is approximated by AR with a sufficiently high order p.
- Which plot: evaluate each PSD at $\theta = 0$ (cosines = 1) and π (alternate signs); AR and MA fitted to the same two lags nearly overlap; a parametric PSD is never negative. *Apr25 3g: corr 1.125 / 2.125, AR 1.56 / 2.57, MA 1.5 / 2.5 → plot D (option a). Apr24 3f: corr 0.06 / 5.06, AR 0.69 / 5.6 → plot B (option c).*

PART 4 · Detection

U16 · Likelihood ratio test → threshold on x

IF YOU SEE "construct the likelihood-ratio test for a threshold η", "show that the LRT takes the form $\sum|x_n| \gtrless \gamma'$", "for which x should H_1 be accepted", "derive the log-likelihood ratio test":

1. $L(x) = \dfrac{p(x; H_1)}{p(x; H_0)}$, H_1 on top. N samples: products → c^N and $\exp(\sum)$.
2. Cancel common factors (x, k_1); $e^a/e^b = e^{a-b}$; $x^a/x^b = x^{a-b}$.
3. $L \gtrless \gamma$, take ln (direction kept). 4. Isolate the data term; dividing by a negative flips the inequality AND swaps the H_0/H_1 labels. 5. $x^2 > c$ with $x > 0 \to x > \sqrt{c}$; $e^x > c \to x > \ln c$. 6. Say "decide H_1 when …"; the answer is a range. 7. Test a far point.

Rayleigh(σ_0) *vs Rayleigh*(σ_1) *(Jun26 4b):* $L = \frac{\sigma_0^2}{\sigma_1^2}\exp[\frac{x^2}{2}(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2})]$ *(bracket* > 0*); decide* H_1 *when* $x > \sqrt{\frac{2\sigma_0^2\sigma_1^2}{\sigma_1^2 - \sigma_0^2}\ln(\eta\sigma_1^2/\sigma_0^2)}$*. Closer* σ*'s →curves overlap →* $P_D \to P_{FA}$*, ROC→ diagonal. Apr25 4a,* $p = k_1\theta x^{k_2\theta-1}$*,* $\theta = 3$ *vs* 6*,* $\lambda = 0.2$*:* $L = \frac{6x^{11}}{3x^5} = 2x^6 \geq 0.2 \Rightarrow x \geq 0.1^{1/6} = 0.68$*. Laplacian (Nov22 4c)* $\frac{1}{2}e^{-|x|}$ *vs* $e^{-2|x|}$*:* $L = 2^N e^{-\sum|x_n|}$*;* $\sum|x_n| \underset{H_1}{\overset{H_0}{\gtrless}} N\ln 2 - \ln\gamma$ *(small sum → H*$_1$*). Gaussian variances (Nov19 4c, Oct21 4b):* $\ln L = N\ln\frac{\sigma_0}{\sigma_1} - \frac{\sum x^2}{2\sigma_1^2} + \frac{\sum x^2}{2\sigma_0^2} > \ln\gamma$*. Rayleigh 1 vs* $\frac{1}{4}$ *(deck):* $L = \frac{1}{4}e^{3x^2/8}$*,* $x > \sqrt{\frac{8}{3}\ln 4\lambda} = 1.92$. Definitions (Jan19 3a): $P_{fa} = \int_{L>\gamma} p(x; H_0)dx$, $P_d = \int_{L>\gamma} p(x; H_1)dx$.

U17 · P_{FA}, P_D, P_M: shade and compute

IF YOU SEE "sketch the likelihoods … indicate P_{FA} and P_M … calculate P_D and P_{FA} for $\gamma = 1$", "for $P_{FA} = 0.1$ calculate the threshold", "plot the ROC":

Curve = the truth, side = what you said. $P_{FA} = H_0$ curve on the H_1 side. $P_D = H_1$ curve on the H_1 side. $P_M = 1 - P_D = H_1$ curve on the H_0 side.

1. $\gamma = 1$: threshold on the axis = where the curves cross: set $p_1 = p_0$, divide, take ln, solve (quadratic if Gaussian vs exponential).
2. Which side is H_1: test $y = 0$ (or a large y); the side where $p_1 > \gamma p_0$. It can be the LEFT (Nov22, Jan20). "Pay attention to the inequality" = this.
3. Integrals with explicit limits; two-sided region $|y| > y_{th}$: double one tail.
4. Evaluate: $\int_0^\gamma \lambda e^{-\lambda y}dy = 1 - e^{-\lambda\gamma}$; $\int_\gamma^\infty \lambda e^{-\lambda y}dy = e^{-\lambda\gamma}$; Gaussian: standardise, $P(X > a) = Q(\frac{a-\mu}{\sigma})$, area to the left $= 1 - Q$; erf table: left area $= \frac{1}{2}(1 + \text{erf})$, right $= \frac{1}{2}(1 - \text{erf})$; band values: tail beyond $3\sigma = 0.5 - 0.341 - 0.136 - 0.021 = 0.002$, beyond $1\sigma = 0.159$.
5. Threshold from a target: write $P_{FA}(\gamma') =$ target, solve with ln. Then P_D at that γ'; reuse $e^{-\gamma'}$: $e^{-2\gamma'} = (e^{-\gamma'})^2$.
6. Check $P_D > P_{FA}$. ROC: P_D vertical, P_{FA} horizontal, through $(0, 0)$ and $(1, 1)$, above the diagonal; more separation → closer to the top-left.

Apr24 4b / Jan22 4b-c: $\frac{1}{4}e^{-y/2} = \frac{1}{\sqrt{2\pi}}e^{-y^2/2} \to e^{y^2/2 - y/2} = \frac{4}{\sqrt{2\pi}} = 1.596 \to y^2 - y - 0.935 = 0 \to y_{th} = 1.588$*;* H_1 *outside* $(|y| > 1.588)$*.* $P_{FA} = 2Q(1.59) = 0.112$*;* $P_D = 2 \cdot \frac{1}{4}\int_{1.588}^\infty e^{-y/2}dy = e^{-0.794} = 0.452$ *(one-sided, as printed in Apr24:* 0.056 *and* 0.226*). Nov22 4d-e,* e^{-y} *vs* $2e^{-2y}$*,* H_1 *when* $y < \gamma'$*:* $P_{FA} = 1 - e^{-\gamma'} = 0.1 \Rightarrow \gamma' = -\ln 0.9 = 0.105$*;* $P_D = 1 - e^{-2\gamma'} = 1 - 0.81 = 0.19$*. Nov19 4b,* $\sigma_1 = 3\sigma_0$*,* $x_{th} = \sigma_1$*:* $P_{fa} = 2(0.002) = 0.004$*,* $P_d = 2(0.159) = 0.318$*. Gaussian means (Oct21 4a):* H_1 *when* $y > y_{th}$*,* $P_{fa} = \frac{1}{2}(1 - \text{erf}(\frac{y_{th}}{\sqrt{2}}))$*,* $P_d = \frac{1}{2}(1 - \text{erf}(\frac{y_{th}-\mu}{\sqrt{2}}))$*. Exponentials (Jan20 5):* H_1 *on the left,* $P_d = 1 - e^{-\lambda_1 x_{th}}$*,* $P_{fa} = 1 - e^{-\lambda_0 x_{th}}$.

U18 · Bayesian detector: priors and costs

IF YOU SEE "prior probabilities $P(H_0) = 0.9$, $P(H_1) = 0.1$ … equal costs … when should H_1 be accepted", "$C_{0,1} = 100C_{1,0}$ … the posterior of H_0 is 5 times smaller", "find γ'' where $p(y|H_1)P_1 = p(y|H_0)P_0$":

C_{ij} = cost of deciding H_i when H_j is true: C_{10} false alarm, C_{01} miss. $L(x) \gtrless \eta = \dfrac{P_0(C_{10} - C_{00})}{P_1(C_{01} - C_{11})}$; equivalently $\dfrac{p(H_1 \mid x)}{p(H_0 \mid x)} \gtrless \dfrac{C_{10} - C_{00}}{C_{01} - C_{11}}$ (posteriors already contain the priors). Equal costs: $\eta = P_0/P_1$ (minimises $P_e = P_0P_{FA} + P_1P_M$). Equal costs and priors: $\eta = 1$ (ML). Scaling all costs changes nothing. Larger P_0 or $C_{10} \to$ larger η; larger P_1 or $C_{01} \to$ smaller η. *Jun26 4c:* $\eta = 0.9/0.1 = 9 \to$ *plug into the Rayleigh LRT:* $x \geq \sqrt{\frac{2\sigma_0^2\sigma_1^2}{\sigma_1^2 - \sigma_0^2}\ln\frac{9\sigma_1^2}{\sigma_0^2}}$*. Apr25 4c: threshold* $C_{10}/(100C_{10}) = 0.01$*; posterior ratio* $5 \geq 0.01 \to H_1$*: the student fails. Nov22 4f-g:* $1.2e^{-2y} = 0.4e^{-y} \Rightarrow e^y = 3$*,* $\gamma'' = \ln 3$*,* H_1 *when* $y < \ln 3$*;* $P_{FA} = 1 - e^{-\ln 3} = \frac{2}{3}$*,* $P_M = e^{-2\ln 3} = \frac{1}{9}$*,* $P_e = 0.4 \cdot \frac{2}{3} + 0.6 \cdot \frac{1}{9} = \frac{1}{3}$.

U19 · Detection MC

Decide H_1, truth H_0: false alarm = false positive = type I (P_{FA}). Decide H_0, truth H_1: miss = false negative = type II (P_M). Neyman–Pearson: for a FIXED P_{FA}, P_D is MAXIMISED (lies: "minimised", "fixed P_D", "$P_{FA} = P_D$"); always a threshold on the likelihood ratio. Bayesian: Bayes risk MINIMISED; $C_{01} = C_{10}$ (1 or 10), $C_{00} = C_{11} = 0$ minimises P_e. Matched filter: maximises output SNR; $h[n] = s[N - 1 - n]$ (time-reversed, not $s[n]$); correlated noise → prewhiten, so "only uncorrelated noise" is false. ROC below the diagonal = swapped hypotheses or axes, or CDF used for P_d; a large H_0 variance cannot cause it (Oct21 1e).

Calculations · the algebra under every answer

- ln: $\ln(ab) = \ln a + \ln b$; $\ln a^n = n\ln a$; $\ln e^u = u$; $\ln\prod = \sum\ln$; $e^a e^b = e^{a+b}$; $e^a/e^b = e^{a-b}$; $(e^a)^n = e^{an}$; $e^{-2\ln 3} = 3^{-2} = \frac{1}{9}$; $e^{-\ln 3} = \frac{1}{3}$; solve $e^{ku} = c$: $u = \ln c/k$.
- Derivatives: $(\ln\theta)' = 1/\theta$; $(\ln(1-\theta))' = -1/(1-\theta)$; $(c/\theta)' = -c/\theta^2$; $(c/\theta^2)' = -2c/\theta^3$; $(e^{a\theta})' = ae^{a\theta}$; $(\ln f)' = f'/f$.
- Integrals: $\int e^{ax} = \frac{1}{a}e^{ax}$; $\int xe^{ax} = (\frac{x}{a} - \frac{1}{a^2})e^{ax}$; $\int x^2e^{ax} = (\frac{x^2}{a} - \frac{2x}{a^2} + \frac{2}{a^3})e^{ax}$; $\int_0^\infty x^n e^{-x}dx = n!$ ($\int_0^\infty x^2e^{-x} = 2$, $\int_0^\infty xe^{-x} = 1$); $\int_0^\infty \theta e^{-c\theta}d\theta = 1/c^2$; $\int_0^\infty \theta^2 e^{-c\theta}d\theta = 2/c^3$; $[F]_a^b = F(b) - F(a)$; at ∞, polynomial $\times e^{-cx} \to 0$; $\int_0^1 y\,dy = \frac{1}{2}$, $\int_0^1 y^2dy = \frac{1}{3}$. Substitution $y = c\theta$: $d\theta = dy/c$.
- Rayleigh moment: $E[x^2] = \int_0^\infty x^2\frac{x}{\theta^2}e^{-x^2/2\theta^2}dx = 2\theta^2$ (put $u = x^2/2\theta^2$). $E[X^2] = \text{Var} + \mu^2$.
- Cosines: $e^{jk\theta} + e^{-jk\theta} = 2\cos k\theta$; $|1 + ae^{-j\theta}|^2 = 1 + a^2 + 2a\cos\theta$; $|1 + a_1e^{-j\theta} + a_2e^{-j2\theta}|^2 = (1 + a_1^2 + a_2^2) + 2(a_1 + a_1a_2)\cos\theta + 2a_2\cos 2\theta$; $\cos\theta = \frac{z+z^{-1}}{2}$; $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$; $\sin^2 = \frac{1-\cos 2\alpha}{2}$; $\cos^2 = \frac{1+\cos 2\alpha}{2}$; at $\theta = \pi$: $\cos k\pi = (-1)^k$; at $\pi/2$: $0, -1, 0, 1$.
- Quadratic $Ax^2 + Bx + C = 0$: $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$; $b^2 + 8b + 1 = 0$: $-4 \pm \sqrt{15}$; $y^2 - y - 0.935 = 0$: $\frac{1 \pm 2.177}{2}$.
- Geometric: $\sum_{n\geq 0} q^n = \frac{1}{1-q}$ ($|q| < 1$).
- Inequalities: ln keeps the direction; multiplying by a negative flips it (and swaps H_0/H_1); $\sqrt{\cdot}$ keeps it for positives.

Numbers

$e^{-1} = 0.3679$, $e^{-2} = 0.1353$, $e^{-0.5} = 0.6065$, $e^{-0.794} = 0.452$, $\ln 2 = 0.693$, $\ln 3 = 1.099$, $\ln 4 = 1.386$, $\ln 10 = 2.303$, $\ln 20 = 2.996$, $\ln 0.9 = -0.1054$, $\ln 0.1 = -2.303$, $\sqrt{2\pi} = 2.5066$, $1/\sqrt{2\pi} = 0.3989$, $4/\sqrt{2\pi} = 1.596$, $\sqrt{15} = 3.873$, $1/(1 - e^{-2}) = 1.1565$, $0.1^{1/6} = 0.681$. $Q(x)$ = area right of x for $\mathcal{N}(0, 1)$; $Q(-x) = 1 - Q(x)$; $Q(0) = 0.5$, $Q(1) = 0.1587$, $Q(1.28) = 0.100$, $Q(1.5) = 0.0668$, $Q(1.59) = 0.0559$, $Q(1.645) = 0.050$, $Q(1.96) = 0.025$, $Q(2) = 0.0228$, $Q(3) = 0.00135$. erf$(x/\sqrt{2})$, $x = 1, 2, 3$: $0.683, 0.954, 0.997$ ($\frac{1}{2}(1 - 0.683) = 0.1585 = Q(1)$).